

1. NAME OF THE MEDICINAL PRODUCT

Desliq ER 50mg Tablet
Desliq ER 100mg Tablet

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Desvenlafaxine Succinate Monohydrate 75.87 mg is equivalent to Desvenlafaxine 50mg
Desvenlafaxine Succinate Monohydrate 151.77mg is equivalent to Desvenlafaxine 100mg

3. PHARMACEUTICAL FORM

Desliq ER 50mg Tablet
Light pink colored, round, biconvex, film coated tablets, debossed with '50' on one side and plain on other side

Desliq ER 100mg Tablet
Dark brown to red colored, round, biconvex film coated tablets, debossed with '100' on one side and plain on other side

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Desvenlafaxine succinate, a selective serotonin and norepinephrine reuptake inhibitor (SNRI), is indicated for the treatment of major depressive disorder (MDD). The efficacy of Desvenlafaxine succinate has been established in four 8-week, placebo controlled studies of outpatients who met DSM-IV criteria for major depressive disorder.

A major depressive episode (DSM-IV) implies a prominent and relatively persistent (nearly every day for at least 2 weeks) depressed or dysphoric mood that usually interferes with daily functioning, and includes at least 5 of the following 9 symptoms: depressed mood, loss of interest in usual activities, significant change in weight and/or appetite, insomnia or hypersomnia, psychomotor agitation or retardation, increased fatigue, feelings of guilt or worthlessness, slowed thinking or impaired concentration, or a suicide attempt or suicidal ideation.

The efficacy of Desvenlafaxine succinate in maintaining a response in major depressive disorder for up to 24 weeks following 12 weeks of acute treatment was demonstrated in a placebo-controlled trial. Nevertheless, the physician who elects to use Desvenlafaxine succinate for extended periods, i.e., beyond 9 months, should continue to periodically re-evaluate the long-term usefulness of the drug for the individual patients.

4.2 Posology and method of administration

Major depressive disorder

The recommended dose for Desvenlafaxine succinate is 50 mg once daily, with or without food. Based on clinical judgement, if dose increases are indicated for individual patient, they should occur gradually and at intervals of not less than 7 days. The maximum dose should not exceed 200 mg/day.

When discontinuing therapy, gradual dose reduction is recommended whenever possible to minimize discontinuation symptoms (**see section: SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

Desvenlafaxine succinate should be taken at approximately the same time each day. Tablets must be swallowed whole with fluid and not divided, crushed, chewed, or dissolved.

Use in patients with renal impairment

The recommended starting dose in patients with severe renal impairment (24-hr CrCl <30 mL/min) or end-stage renal disease (ESRD) is 50 mg every other day. Because of individual variability in clearance in these patients, individualization of dosage may be desirable. Supplemental doses should not be given to patients after dialysis.

Use in patients with hepatic impairment

No dosage adjustment is necessary for patients with hepatic impairment.

Pediatric use

Safety and effectiveness in patients less than 18 years of age have not been established.

Use in elderly patients

No dosage adjustment is required solely on the basis of age; however, possible reduced renal clearance of desvenlafaxine should be considered when determining dose (**see section 5.2: PHARMACOKINETIC PROPERTIES**).

Greater sensitivity to desvenlafaxine in some older patients cannot be ruled out.

Discontinuing desvenlafaxine

Symptoms associated with discontinuation of desvenlafaxine, other SNRIs and SSRIs have been reported. Patients should be monitored for these symptoms when discontinuing treatment. A gradual reduction in the dose rather than abrupt cessation is recommended. If intolerable symptoms occur following a decrease in the dose or upon discontinuation of treatment, then resuming the previously prescribed dose may be considered. Subsequently, the physician may continue decreasing the dose, but at a more gradual rate (**see section 4.8: UNDESIRABLE EFFECTS**). In some patients, discontinuation may need to occur over periods of months or longer.

Switching patients from other antidepressants to desvenlafaxine

Discontinuation symptoms have been reported when switching patients from other antidepressants, including venlafaxine, to desvenlafaxine. Tapering of the initial antidepressant may be necessary to minimize discontinuation symptoms.

Use of desvenlafaxine with reversible MAOIs such as linezolid or methylene blue

Do not start Desvenlafaxine succinate in a patient who is being treated with a reversible MAOI such as linezolid or in whom intravenous methylene blue has been administered because there is increased risk of serotonin syndrome. In a patient who requires more urgent treatment of a psychiatric condition, non-pharmacological interventions, including hospitalization, should be considered (**see section 4.3: CONTRAINDICATIONS**).

In some cases, a patient already receiving Desvenlafaxine succinate therapy may require urgent treatment with linezolid or intravenous methylene blue. If acceptable alternatives to linezolid or intravenous methylene blue treatment are not available and the potential benefits of linezolid or intravenous methylene blue treatment are judged to outweigh the risks of serotonin syndrome in a particular patient, Desvenlafaxine succinate should be stopped promptly, and linezolid or intravenous methylene blue can be administered. The patient should be monitored for symptoms of serotonin syndrome for two weeks or until 24 hours after the last dose of linezolid or intravenous methylene blue, whichever comes first. Therapy with Desvenlafaxine succinate may be resumed 24 hours after the last dose of linezolid or intravenous methylene blue (**see section 4.4: SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

The risk of administering methylene blue by non-intravenous routes (such as oral tablets or by local injection) or in intravenous doses much lower than 1 mg/kg with Desvenlafaxine succinate is unclear. The clinician should, nevertheless, be aware of the possibility of emergent symptoms of serotonin syndrome with such use (**see section 4.4: SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

4.3 Contraindications

Hypersensitivity to desvenlafaxine succinate, venlafaxine hydrochloride or to any excipients in the desvenlafaxine formulation.

Desvenlafaxine is an inhibitor of both norepinephrine and serotonin reuptake. Desvenlafaxine succinate must not be used in combination with a monoamine oxidase inhibitor (MAOI), or within at least 14 days of discontinuing treatment with an MAOI. Based on the half-life of desvenlafaxine succinate, at least 7 days should be allowed after stopping desvenlafaxine succinate before starting an MAOI. Starting desvenlafaxine succinate in a patient who is being treated with a reversible MAOI such as linezolid or in whom intravenous methylene blue has been administered is also contraindicated because of an increased risk of serotonin syndrome (**see sections 4.2 and 4.4: POSOLOGY AND METHOD OF ADMINISTRATION; SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

4.4 Special warnings and precautions for use

Special Warnings

Clinical worsening of depressive symptoms, unusual changes in behavior, and suicidality

Desvenlafaxine succinate is an SNRI, a class of medicines that may be used to treat depression. All patients treated with desvenlafaxine should be monitored appropriately and observed closely for clinical worsening, suicidality, or unusual change in behaviour. They, their families, and their caregivers should be encouraged to be alert to the emergence of anxiety, agitation, panic attacks, insomnia, irritability, hostility, aggressiveness, impulsivity, akathisia (psychomotor restlessness), hypomania, mania, other unusual changes in behavior, worsening of depression, and suicidal ideation, especially when initiating therapy or during any change in dose or dosage regimen. The risk of suicide attempt must be considered, especially in depressed, and the smallest quantity of drug, consistent with good patients management, should be provided to reduce the risk of overdose.

Suicidality in Children and Adolescents

Antidepressants increase the risk of suicidal thinking and behavior (suicidality) in children and adolescents with major depressive disorder (MDD) and other psychiatric disorders.

Anyone considering the use of an antidepressant in a child or adolescent for any clinical use must balance the risk of increased suicidality with the clinical need. Patients who are started on therapy should be observed closely for clinical worsening, suicidality, or unusual changes in behavior.

Families and caregivers should be advised to closely observe the patient and to communicate with the prescriber.

Discontinuation effects

During marketing of serotonin-norepinephrine reuptake inhibitors (SNRIs) and selective serotonin reuptake inhibitors (SSRIs), there have been post-marketing reports of adverse events occurring upon discontinuation of these drugs, particularly when abrupt, including the following: dysphoric mood, irritability, agitation, dizziness, sensory disturbances (e.g., paresthesias such as electric shock sensations), anxiety, confusion, headache, lethargy, emotional lability, insomnia, hypomania, tinnitus, seizures, visual impairment, and hypertension.

While these events are generally self-limiting, there have been reports of serious discontinuation symptoms, and sometimes these effects can be protracted and severe. In addition, suicide/suicidal thoughts and aggression have been observed in patients during changes in desvenlafaxine dosing regimen, including during discontinuation.

Patients should be monitored when discontinuing treatment with desvenlafaxine. A gradual reduction in the dose rather than abrupt cessation is recommended whenever possible. If intolerable symptoms occur following a decrease in the dose or upon discontinuation of treatment, then resuming the previously prescribed dose may be considered (**see sections 4.2 and 4.8: POSOLOGY AND METHOD OF ADMINISTRATION; UNDESIRABLE EFFECTS**).

Sexual dysfunction

Serotonin-norepinephrine reuptake inhibitors (SNRIs) may cause symptoms of sexual dysfunction (**see section 4.8: UNDESIRABLE EFFECTS**). There have been reports of long-lasting sexual dysfunction where the symptoms have continued despite discontinuation of SNRIs.

Mania/Hypomania

Activation of mania/hypomania has also been reported in a small proportion with major affective disorder who were treated with other marketed antidepressants. As with all antidepressants, desvenlafaxine should be used cautiously in patients with a history or family history of mania or hypomania (**see section 4.8: UNDESIRABLE EFFECTS**).

Serotonin syndrome or Neuroleptic Malignant Syndrome (NMS)-like reactions

As with other serotonergic agents, the development of a potentially life-threatening serotonin syndrome or Neuroleptic Malignant Syndrome (NMS)-like reactions may occur with desvenlafaxine treatment, particularly with concomitant use of other serotonergic drugs (including SSRIs, SNRIs, amphetamines and triptans), with drugs that impair metabolism of serotonin (e.g., MAOIs, including reversible MAOIs such as linezolid and intravenous methylene blue), or with antipsychotics or other dopamine antagonists (see sections 4.2 and 4.3: POSOLOGY AND METHOD OF ADMINISTRATION; CONTRAINDICATIONS). Serotonin syndrome symptoms may include mental status changes (agitation, hallucinations, and coma), autonomic instability (e.g., tachycardia, labile blood pressure, and hyperthermia), neuromuscular aberrations (e.g., hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g., nausea, vomiting, and diarrhea). Serotonin syndrome, in its most severe form, can resemble NMS, which includes hyperthermia, muscle rigidity, autonomic instability with possible rapid fluctuation of vital signs, and mental status changes (see section 4.5: INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION). If concomitant treatment with desvenlafaxine and other agents that may affect the serotonergic and/or dopaminergic neurotransmitter systems is clinically warranted, careful observation of the patient is advised, particularly during treatment initiation and dose increases. The concomitant use of desvenlafaxine with serotonin precursors (such as tryptophan supplements) is not recommended.

Narrow-angle glaucoma

Mydriasis has been reported in association with desvenlafaxine; therefore, patients with raised intraocular pressure or those at risk of acute narrow-angle glaucoma (angle-closure glaucoma) should be monitored (**see section 4.8: UNDESIRABLE EFFECTS**).

Other information - Residual inert matrix tablet

Patients receiving desvenlafaxine succinate may notice an inert matrix tablet passing in the stool or via colostomy. Patients should be informed that the active medication has already been absorbed by the time the patient sees the inert matrix tablet.

Precautions

Co-administration of drugs containing venlafaxine and/or desvenlafaxine

Desvenlafaxine is the major active metabolite of venlafaxine. Products containing desvenlafaxine succinate should not be used concomitantly with products containing venlafaxine hydrochloride or other products containing desvenlafaxine succinate.

Effects on blood pressure

Pre-existing hypertension should be controlled before treatment with desvenlafaxine. Patients receiving desvenlafaxine should have regular monitoring of blood pressure. Cases of elevated blood pressure requiring immediate treatment have been reported with desvenlafaxine.

Sustained blood pressure increases could have adverse consequences. For patients who experience a sustained increase in blood pressure while receiving desvenlafaxine, either dose reduction or discontinuation should be considered. Caution should be exercised in treating patients with underlying conditions that might be compromised by increases in blood pressure (**see section 4.8: UNDESIRABLE EFFECTS**).

Cardiovascular/cerebrovascular disorders

Caution is advised in administering desvenlafaxine to patients with cardiovascular, cerebrovascular, or lipid metabolism disorders. Desvenlafaxine has not been evaluated systematically in patients with a recent history of myocardial infarction, unstable heart disease, uncontrolled hypertension, or cerebrovascular disease. (**see section 4.8: UNDESIRABLE EFFECTS**).

Serum lipids

Measurement of serum lipids should be considered during treatment with desvenlafaxine (**see section 4.8: UNDESIRABLE EFFECTS**).

Seizures

Desvenlafaxine has not been systematically evaluated with a seizure disorder. Desvenlafaxine should be prescribed with caution in patients who have a seizure disorder (**see section 4.8: UNDESIRABLE EFFECTS**).

Abnormal bleeding

Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs), including desvenlafaxine, may increase risk of bleeding events. Concomitant use of aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs), warfarin, and other anticoagulants may add to this risk. Bleeding events related to SSRIs and SNRIs have ranged from ecchymosis, hematoma, epistaxis, and petechiae to life-threatening hemorrhages. Patients should be cautioned about the risk of bleeding associated with the concomitant use of desvenlafaxine and NSAIDs, aspirin, or other drugs that affect coagulation or bleeding.

Hyponatremia

Cases of hyponatremia and/or the Syndrome of Inappropriate Antidiuretic Hormone (SIADH) secretion have been described with SNRIs (including desvenlafaxine succinate) and SSRIs, usually in volume-depleted or dehydrated, including elderly patients and patients taking diuretics (**see section 4.8: UNDESIRABLE EFFECTS**).

4.5 Interaction with other medicinal products and other forms of interaction

Monoamine oxidase inhibitors (MAOI)

Adverse reactions, some of which were serious, have been reported in patients who have recently been discontinued from a monoamine oxidase inhibitor (including reversible MAOIs such as linezolid and intravenous methylene blue) and started on antidepressants with pharmacological properties similar to desvenlafaxine (SNRIs or SSRIs), or who have recently had SNRI or SSRI therapy discontinued prior to initiation of an MAOI (**see sections 4.2 and 4.4: POSOLOGY AND METHOD OF ADMINISTRATION; SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

Concomitant use of desvenlafaxine in patients taking monoamine oxidase inhibitors (MAOIs) is contraindicated (**see section 4.3: CONTRAINDICATIONS**)

Central nervous system (CNS)-active agents

The risk of using desvenlafaxine in combination with other CNS-active drugs has not been systematically evaluated. Consequently, caution is advised when desvenlafaxine is taken in combination with other CNS-active drugs.

Serotonin syndrome

As with other serotonergic agents, serotonin syndrome, a potentially life-threatening condition, may occur with desvenlafaxine treatment, particularly with concomitant use of other agents that may affect the serotonergic neurotransmitter system (including triptans, SSRIs, other SNRIs, amphetamines, lithium, sibutramine, fentanyl and its analogues, tramadol, dextromethorphan, tapentadol, meperidine, methadone, pentazocine, or St. John's Wort [*Hypericum perforatum*]), with drugs that impair metabolism of serotonin (such as MAOIs, including linezolid [an antibiotic which is a reversible non-selective MAOI] and methylene blue), or with serotonin precursors (such as tryptophan supplements) (**see sections 4.2, 4.3 and 4.4: POSOLOGY AND METHOD OF ADMINISTRATION; CONTRAINDICATIONS; SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

If concomitant treatment with desvenlafaxine and an SSRI, an SNRI or a 5-hydroxytryptamine receptor agonist (triptan) is clinically warranted, careful observation of the patient is advised, particularly during treatment initiation and dose increases. The concomitant use of desvenlafaxine with serotonin precursors (such as tryptophan supplements) is not recommended (**see section 4.4: SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

Ethanol

Desvenlafaxine does not increase the impairment of mental and motor skills caused by ethanol. However, as with all CNS-active drugs, patients should be advised to avoid alcohol consumption while taking desvenlafaxine.

Potential for desvenlafaxine to affect other drugs

Inhibitors of CYP3A4

CYP3A4 is minimally involved in desvenlafaxine elimination. Ketoconazole (200 mg BID) increased the area under the concentration vs. time curve (AUC) of desvenlafaxine (400 mg single dose) by approximately 43%, a weak interaction, and Cmax by about 8%. Concomitant use of desvenlafaxine with potent inhibitors of CYP3A4 may result in higher exposure to desvenlafaxine.

Inhibitors of other CYP enzymes

Drugs that inhibit CYP isozymes 1A1, 1A2, 2A6, 2D6, 2C8, 2C9, 2C19, and 2E1 are not expected to have significant impact on the pharmacokinetic profile of desvenlafaxine.

480 mm

150 mm
Front Side

Drugs metabolized by CYP2D6

When desvenlafaxine succinate was administered at a dose of 100 mg daily in conjunction with a single 50 mg dose of desipramine, a CYP2D6 substrate, the AUC of desipramine increased approximately 17%. When 400 mg was administered, the AUC of desipramine increased approximately 90%. When desvenlafaxine succinate was administered at a dose of 100 mg daily in conjunction with a single 60 mg dose of codeine, a CYP2D6 substrate metabolized to morphine, the AUC of codeine was unchanged, the AUC of morphine decreased approximately 8%. Concomitant use of desvenlafaxine with a drug metabolized by CYP2D6 may result in increased concentrations of that drug and decreased concentrations of its CYP2D6 metabolites.

4.6 Fertility, pregnancy and lactation

Pregnancy

The safety of desvenlafaxine in human pregnancy has not been established. It is demonstrated that desvenlafaxine crosses the human placenta. Desvenlafaxine must only be administered to pregnant women if the expected benefits outweigh the possible risks. If desvenlafaxine is used until, or shortly before birth, discontinuation effects in the newborn should be considered. It is demonstrated that desvenlafaxine crosses the placenta.

Exposure to SNRIs in mid to late pregnancy may increase the risk for preclampsia, and exposure to SNRIs near delivery may increase the risk for postpartum hemorrhage.

Lactation

Desvenlafaxine (O-desmethylvenlafaxine) is excreted in human milk. No adverse events occurred in either the lactating mothers or the nursing infants, however, the effect in infants have not been established. Desvenlafaxine should only be taken by lactating women if the expected benefits outweigh the possible risks.

Pregnancy and lactation

Observational data indicate an increased risk (less than 2-fold) of postpartum haemorrhage following SSRI/SNRI exposure within the month prior to birth.

4.7 Effects on ability to drive and use machines

Interference with cognitive and motor performance

Any CNS-active drug may impair judgment, thinking, or motor skills, patients should be cautioned about operating hazardous machinery, including automobiles, until they are reasonably certain that desvenlafaxine therapy does not adversely affect their ability to engage in such activities.

4.8 Undesirable effects

ADRs by SOC and CIOMS frequency category listed in order of decreasing medical seriousness within each frequency category and SOC.

Adverse Events Table - Combined MDD and VMS Studies						
System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare	Frequency Not Known (cannot be estimated from available data)
Immune System Disorders			Hypersensitivity			
Metabolism and Nutrition Disorders		Decreased appetite		Hyponatremia		
Psychiatric Disorders	Insomnia	Withdrawal syndrome, anxiety, nervousness, abnormal dreams, irritability, libido decreased, anorgasmia	Depersonalisation, orgasm abnormal	Mania, hypomania, hallucination		
Nervous System Disorders	Headache, dizziness, somnolence	Tremor, paraesthesia, disturbance in attention, dysgeusia	Syncope, extrapyramidal disorder	Serotonin syndrome*†, convulsion		
Eye Disorders		Vision blurred, mydriasis				
Ear and Labyrinth Disorders		Vertigo, tinnitus				
Cardiac Disorders		Tachycardia, palpitations		Stress cardiomyopathy (takotsubo cardiomyopathy)*†		
Vascular Disorders		Blood pressure increased, hot flush	Orthostatic hypotension, peripheral coldness			
Respiratory, Thoracic, and Mediastinal Disorders		Yawning	Epistaxis			
Gastrointestinal Disorders	Nausea, dry mouth, constipation	Diarrhoea, vomiting		Pancreatitis acute		
Skin and Subcutaneous Tissue Disorders		Hyperhidrosis, rash	Alopecia	Stevens-Johnson syndrome*†, angioedema, photosensitivity reaction		
Musculoskeletal and Connective Tissue and Disorders		Musculoskeletal stiffness				
Renal and Urinary Disorders			Urinary retention, urinary hesitation, proteinuria			
Reproductive System and Breast Disorders		Erectile dysfunction # ejaculation delayed #	Ejaculation disorder*# Ejaculation failure‡, sexual dysfunction			
General Disorders and Administration Site Conditions		Fatigue, asthenia, chills, feeling jittery				
Investigations		Liver function test abnormal, weight increased, weight decreased	Blood cholesterol increased, blood triglycerides increased, blood prolactin increased			
* ADR identified post-marketing # ADR Frequency estimated using "The Rule of 3" ‡ Frequency is calculated based on men only ADR = Adverse drug reaction; MDD = Major depressive disorder; VMS = Vasomotor symptoms						

Ischemic cardiac adverse events

There were uncommon reports of ischemic cardiac adverse events, including myocardial ischemia, myocardial infarction, and coronary occlusion requiring revascularization; these patients had multiple underlying cardiac risk factors. More patients experienced these events during desvenlafaxine treatment as compared to placebo (**see section 4.4: SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

Discontinuation symptoms

Adverse drug reactions reported in association with abrupt discontinuation, dose reduction or tapering of treatment in MDD at a rate of ≥2% include: dizziness, withdrawal syndrome, nausea and headache. In general, discontinuation symptoms occurred more frequently with higher doses and with longer duration of therapy (**see sections 4.2 and 4.4: POSOLOGY AND METHOD OF ADMINISTRATION; SPECIAL WARNINGS AND PRECAUTIONS FOR USE**).

Adverse reactions

Pediatric patients

In general, the adverse reaction profile of Desvenlafaxine Succinate Extended Release Tablets in children and adolescents (aged 7 to 17) was similar to that seen for adults.

The most frequently reported events were: Headache, Nausea, Abdominal pain upper, Nasopharyngitis, Dizziness, Upper respiratory tract infection, Decreased appetite, Vomiting, Fatigue, and Insomnia.

When compared to adverse event rates in adults, the following events occurred more frequently in pediatric patients: Abdominal pain upper, Weight increased, Gastroenteritis viral, Dysmenorrhoea, Accidental overdose, Cough, Irritability, Oropharyngeal pain, and Sinusitis.

Geriatric use

No overall differences in safety or efficacy were observed between these patients and younger patients; however, there was a higher incidence of systolic orthostatic hypotension and increases in systolic blood pressure in patients ≥65 years of age compared to patients <65 years of age treated with desvenlafaxine.

Adverse reactions reported with other SNRIs

Although gastrointestinal bleeding is not considered an adverse reaction for desvenlafaxine succinate, it is an adverse reaction for other SNRIs and may also occur with desvenlafaxine succinate.

4.9 Overdose

Among the patients included in the major depressive disorder clinical trials of desvenlafaxine succinate, there were four adults who ingested doses greater than 800 mg of desvenlafaxine succinate (4,000 mg [desvenlafaxine alone], 900, 1,800 and 5,200 mg [in combination with other drugs]); all patients recovered. In addition, one patient's 11-month-old child accidentally ingested 600 mg of desvenlafaxine succinate, was treated, and recovered.

No specific antidotes for desvenlafaxine are known. Induction of emesis is not recommended. Because of the moderate volume of distribution of this drug, forced diuresis, dialysis, hemoperfusion, and exchange transfusion are unlikely to be of benefit.

Treatment should consist of those general measures employed in the management of overdose with any SSRI/SNRI. Ensure an adequate airway, oxygenation, and ventilation. Monitor cardiac rhythm and vital signs. General

supportive and symptomatic measures are also recommended. Gastric lavage with a large-bore orogastric tube with appropriate airway protection, if needed, may be indicated if performed soon after ingestion or in symptomatic patients. Activated charcoal should be administered.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Mode of Action

It has been shown that desvenlafaxine is a selective serotonin and norepinephrine reuptake inhibitor (SNRI). The clinical efficacy of desvenlafaxine is thought to be related to the potentiation of these neurotransmitters in the central nervous system.

Desvenlafaxine lacked significant affinity for numerous receptors, including muscarinic-cholinergic, H1-histaminergic, or α1-adrenergic receptors in vitro. Pharmacologic activity at these receptors has been hypothesized to be associated with the various anticholinergic, sedative, and cardiovascular effects seen with other psychotropic drugs. In the same comprehensive binding profile assay, desvenlafaxine also lacked significant affinity for various ion channels, including calcium, chloride, potassium and sodium ion channels and also lacked monoamine oxidase (MAO) inhibitory activity. Desvenlafaxine lacked significant activity in the in vitro cardiac potassium channel (hERG) assay. It is demonstrated that desvenlafaxine activity is predictive of antidepressant, anxiolytic and thermoregulatory actions, and pain inhibitory properties.

5.2 Pharmacokinetic properties

The single-dose pharmacokinetics of desvenlafaxine are linear and dose-proportional in a dose range of 50 to 600 mg/day. The mean terminal half-life (t1/2), is approximately 11 hours. With once-daily dosing, steady-state plasma concentrations are achieved within approximately 4 to 5 days. At steady-state, multiple-dose accumulation of desvenlafaxine is linear and predictable from the single-dose pharmacokinetic profile.

Absorption and distribution

Desvenlafaxine succinate is well absorbed, with an absolute oral bioavailability of 80%. Mean time to peak plasma concentrations (Tmax) is about 7.5 hours after oral administration. AUC and Cmax of 6,747 ng·hr/mL and 376 ng/mL, respectively, are observed after multiple doses of 100 mg. Under fasting and fed conditions (high-fat meal) indicated that the Cmax was increased about 16% in the fed state, while the AUCs were similar. This difference is not clinically significant; therefore, desvenlafaxine can be taken without regard to meals. The plasma protein binding of desvenlafaxine is low (30%) and is independent of drug concentration. Desvenlafaxine volume of distribution at steady-state following intravenous administration is 3.4 L/kg, indicating distribution into nonvascular compartments.

Metabolism and elimination

Approximately 45% of desvenlafaxine is excreted unchanged in urine. Desvenlafaxine is primarily metabolized by conjugation (mediated by UGT isoforms, including UGT1A1, UGT1A3, UGT2B4, UGT2B15, and UGT2B17) and to a minor extent through oxidative metabolism. Approximately 19% of the administered dose is excreted as the glucuronide metabolite and <5% as the oxidative metabolite (N,O-desmethylvenlafaxine) in urine. CYP3A4 is the predominant cytochrome P450 isozyme mediating the oxidative metabolism (N-demethylation) of desvenlafaxine.

Geriatric

Administration of doses up to 300 mg, saw an age-dependent decrease in desvenlafaxine clearance, resulting in a 32% increase in Cmax and a 55% increase in AUC values for those greater than 75 years of age, as compared with those from 18 to 45 years of age. No dosage adjustment is required solely on the basis of age; however, possible reduced renal clearance of desvenlafaxine should be considered when determining dose.

Hepatic insufficiency

The recommended dose in hepatic impairment is 50 mg/day. Dose escalation above 100 mg/day is not recommended.

Renal insufficiency

The recommended dose in moderate renal impairment is 50 mg per day. Dosage adjustment (50 mg every other day) is recommended in severe renal impairment or ESRD. Doses should not be escalated in moderate or severe renal impairment, or ESRD [See section 8 and See section 13].

6. PHARMACEUTICAL PARTICULARS

6.1 Incompatibilities

Not applicable

6.2 Shelf-life

Please refer to details on outer carton.

6.3 Special precautions for storage

Store below 30°C.

6.4 Nature and contents of container

Alu-Alu (Alu foil-CFB foil) Blister pack of **14 tablets, 2 such blisters in a carton**. Alu-Alu blister made from printed aluminium foil coated with heat seal lacquer (HSL) and cold form blister plain cold form blister film, a laminate made of OPA film layer, soft tempered Aluminium foil and Transparent PVC film. The layers are bonded with adhesive layers.

7. Manufacturer

Alembic Pharmaceuticals Limited
(Formulation Division),
At- Panelav, Tal- Halol, City: Panelav,
Dist: Panchmahal, Gujarat - 389350,
India.

Product Registration Holder

GENPHARMA SDN. BHD.
Lot 5016, Jalan Teratai,
5 ½ Mile Off Jalan Meru,
41050 Klang, Selangor, Malaysia.

Date of Revision: 24 July 2026

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Back Side